

Intro to Alignment Algorithms:

9/15

Global and local

Dayhoff's books shown

Sequence comparison

Global similarity

- scoring schemes
- edit graphs
- alignment = path in edit graph
- principle of optimality
- dynamic programming
- trace back

Sequence Alignment

Two strings x and y over the same alphabet

Input: 2 seqs over Σ and a scoring scheme

Output: an alignment of the 2 sequences

$C \ T \ -$ ← indel
 $\swarrow \quad \uparrow$
 $C \ A \ C$
 match mismatch

Scoring Schemes

Unit Score

δ	A	C	G	T	-
A	1				
C		1			
G			1		
T				1	
-					0

Alignment

$A \ C \ G$
 $| \quad | \quad |$
 $A \ G \ G$

$$\begin{aligned}
 & \delta(A, A) \quad \delta(C, G) \quad \delta(G, G) \\
 \text{score} &= 1 + 0 + 1 = 2
 \end{aligned}$$

Graphs

AAA GGG

GGG AAA

0



- - - AAA GGG

G G G A A A - - -

3

Dayhoff score (for protein sequences)

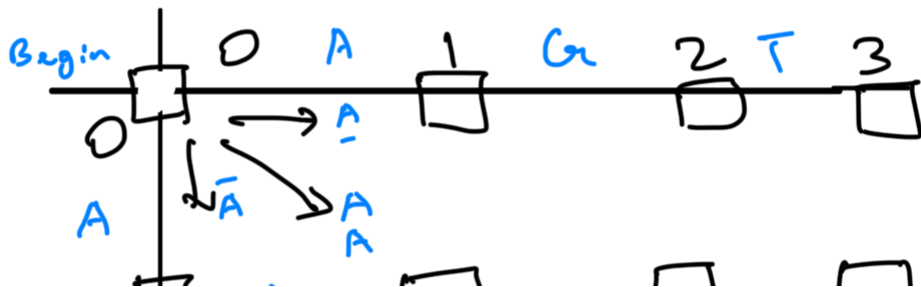
δ	-	A	R	N	D
-					
A			3		
R				6	
N					...
D					

The Edit Graph

$$AGT \rightleftharpoons AT$$

length = 3

length 2



Edit graph

$(3+1) * (2+1)$
nodes



end

Graph is directed

* From any node there is only 3 directions we can go

Alignment = path in the edit graph

AGT
A - T



} every path corresponds to an alignment

The principle of Optimality

→ Optimal ans : in terms of optimal answer for its sub-problems

Dynamic programming

Given : X , Y sequences

Find : optimal alignment w/ best score

1 : find score

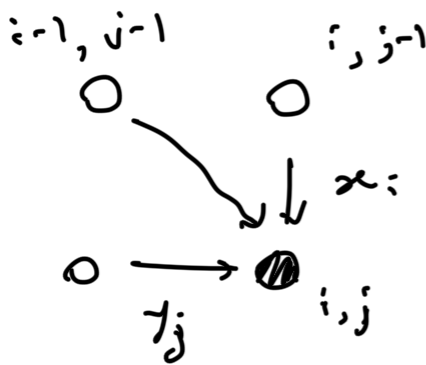
2 : find alignment

2: time

DP matrix

$$\text{Matrix } S = S[S(i, j)]$$

$S(i, j)$ = score of max cost path from beginning vertex and the vertex (i, j)



$$S(i, j) = \text{Max} \begin{cases} S(i-1, j-1) + \delta(x_i, y_j) \\ S(i-1, j) + \delta(x_i, -) \\ S(i, j-1) + \delta(-, y_j) \end{cases}$$

Basic algorithm : local similarity

$$S(i, j) = \text{Max} \begin{cases} 0, \\ S(i-1, j-1) + \delta(x_i, y_j) \\ S(i-1, j) + \delta(x_i, -) \\ S(i, j-1) + \delta(-, y_j) \end{cases}$$

Assumptions :

1. independence of mutations at different sites
* additive scoring scheme
2. gaps of any length $\rightarrow$ one mutation

Board lecture

Heuristic interpretation of the
"Alignment Score"

$\alpha \rightarrow A A C G T A A$

$\gamma \rightarrow A C G G T G T$

A A C G - T A A

4 matches

A - C G G T G T

2 gaps

2 mismatches

let p be the probability of match

$q \rightarrow$ substitution / mismatch

$r \rightarrow$ single letter indel gap

What is the probability of ^{the} particular alignment?

$$\rightarrow p = p^4 q^2 r^2$$

The log-likelihood of this probability

$$\lambda' = \log P_{\text{alignment}} = 4 \log p + 2 \log q + 2 \log r$$

$$\lambda = \lambda' - 8 \log k \quad \text{where } k = \text{constant}$$

$$\begin{aligned} &= 4 \log \left(\frac{p}{k} \right) + 2 \log \left(\frac{q}{k} \right) + 2 \log \left(\frac{r}{k} \right) \\ &= 4 - 2\mu - 2\gamma \end{aligned}$$

* so by looking at likelihood we get intuition of what the scoring scheme could be

$$\left\{ \begin{array}{l} \mu = \log \left(\frac{k}{q} \right) \\ \gamma = \log \left(\frac{k}{r} \right) \\ \text{choosing} \\ k \text{ such that} \\ \log \left(\frac{p}{k} \right) = 1 \end{array} \right.$$